

Advanced SQL

07/04/26

Null functions

- No value / data
- Blank
- No record
- Missing information

Null functions

↳ functions that help us to deal with Null values in our dataset.

Null $\neq$ 0

① Understand if you have missing values in the dataset.

↳ Handle Null values to process data effectively.

fruitsales

sale-id	Product	Amount
1	Banana	100
2	Apple	Null
3	Orange	20

$100 + \text{Null} + 20$
 $\Rightarrow \text{ERROR}$

Select Sum(Amount) As total-revenue
from fruit-sales;

Case 1: How to check for null values in a specific table?

IS NULL

Select columns)

from table-name

WHERE column-name IS Null;

↳ To check for null values

Sale-id	Product	Amount
1	Banana	100
2	Apple	Null
3	Orange	20
4	Grapes	40
5	Lemons	Null

IS NOT Null

Product	Amount
Banana	100
Orange	20
Grapes	40

IS Null

Product	Amount
Apple	Null
Lemons	Null

SELECT product,

amount

FROM

fruit-sales

WHERE

amount **IS Null**;

SELECT product,

amount

FROM fruit-sales

WHERE amount **IS NOT Null**;

IS NOT Null

↳ Rows where there's no Null values

Case 2: We want to find the rows where a specific columns do not have Null values

Syntax:

SELECT column(s)

FROM table-name

WHERE column-name **IS NOT Null**;

Case 3: Replacing a null value on a specific column with a default value

IF NULL → Column-name, replacement-value

Syntax:

SELECT column(s)

IFNULL (column-name, replacement)
new-column-name

AS

FROM table-name;

* Placed on SELECT statement

* Uses 'AS' to give a descriptive name.

SELECT sale-id,
 Product,
 IFNULL (amount, 0) AS amount
 FROM fruit-sales;

Sale-id	Product	amount
1	Banana	100
2	Apple	0
3	Orange	20
4	Grape	40
5	Lemon	0

Case 4: We want to have multiple alternative values

COALESCE (exp1, exp2, exp3) - Giving multiple options /

- Returns the first non-null value from a list of expressions. expression to replace Null
- Checks the expressions from left to right (1)

boxing-team

name	student-id	id-no
Rohi	101	Null
Mark	Null	111
Elon	103	112
Jeff	102	Null
Larry	Null	114
Warren	Null	Null

SELECT name,
 COALESCE (student-id, id-no, 0)
 FROM AS unique-id
 FROM boxing-team;

Name	Unique-id
Rohi	101
Mark	111
Elon	103
Jeff	102
Larry	114
Warren	0

Type of data

Date functions

- Indicates a specific period.
- Indicates an event.
- A specific time when an action was taken.

=> Is a function that help us to ^{Manipulate} handle dates in our dataset. found within the SELECT statement. Creates new column.

What it does: - Extract year,

day,

OLTP database

week,

↳ Captures transactions at

month,

a specific period of

hour,

time.

minutes,

Quarters,

seconds,

① **YEAR()** - used to extract the year on date column.

SELECT YEAR('2025-06-02') AS Year-of-love

FROM table.name;

Year-of-love
2025

② **MONTH()** SELECT MONTH('2025-06-02') AS month-value

month-value
06

③ **DAY()** : SELECT DAY('2025-06-02') AS Day-value

Day-value
02

④ DAYNAME ()

↳ Extract Mon-Sun based on system calendar (Time-zone)

⑤ MONTHNAME ()

↳ Extracts month name (Jan-Dec) based on the system time-zone

SELECT DAYNAME ('2025-06-02') AS Dt
SELECT MONTHNAME ('2025-06-02') AS month-name

Dt
Mon

Month-name
June

⑥ DATEADD (unit / value / date)

↳ It is used to add or subtract date interval
unit - Day, month, Year

Value - Represents increment (increase date)
Decrement (decrease date)

Date - Date column / value

SELECT DATEADD (Day, 2, '2025-06-02') AS 2-later

2-later
2025-06-04

SELECT DATEADD (Day, -2, '2025-06-02') AS 2-day-less

2-day-less
2025-05-31

SELECT DATEADD (Year, 2, '2025-06-02') AS new-date

new-date
2027-06-02

DATEDIFF (unit, date1, date2)

↳ Used to calculate the difference between two dates

$\text{date2} - \text{date1} = \text{value}$

Unit = day, month, year

```
SELECT DATEDIFF (year, '2025-06-02', '2027-06-02')  
[2]
```

CURRENT-DATE ()

↳ Gives today's date based on date management ^{DMS} Sy settings.

e.g.

```
SELECT CURRENT-DATE ( );
```


'2026-04-13'

CURRENT-TIMESTAMP ()

↳ Gives current time based on time-zone.

↳ Gives both date & time.

```
SELECT CURRENT-TIMESTAMP ( );
```

'2026-04-13' 20:05:54.131

